SMARTBRIDGE Let's Bridge the Gap

SKILL WALLET A SmartBridge Product

Project Initialization and Planning Phase

Date	28 June 2026
Team ID	
Project Title	Rising Waters
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform flood forecasting and disaster preparedness using machine learning, boosting prediction accuracy and warning system efficiency. It tackles the inefficiencies of traditional forecasting methods, promising better emergency operations, reduced risks to human lives, and safer communities.

Project Overview	
Objective	The primary objective is to build an intelligent flood prediction system using machine learning algorithms to analyze weather parameters and accurately predict flood risks, delivering early warnings that enable proactive disaster management.
Scope	The project comprehensively develops an environmental forecasting system. It includes collecting weather-related data from open platforms , applying advanced preprocessing and feature engineering , evaluating multiple classification models (Decision Tree, Random Forest, KNN, and XGBoost)
Problem Statement	
Description	Conventional flood forecasting and tracking systems often fail to process complex weather patterns quickly enough to provide timely, localized, and highly accurate warnings. This technological gap directly limits the ability of emergency management authorities
Impact	Failing to solve these system inefficiencies leads to devastating annual losses of local human lives and household assets during recurring monsoon cycles. Furthermore, because accurate risk metrics are not publicly transparent or easily accessible.
Proposed Solution	
Approach	Employing supervised machine learning classification techniques to evaluate critical meteorological inputs—such as annual rainfall, cloud visibility, and seasonal rainfall patterns—to generate instant, data-driven flood risk classifications.
Key Features	1.High-Accuracy Predictive Engine 2. Interactive Early Warning Interface

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel i3 Processor or above
Memory	RAM specifications	Minimum 4 GB RAM
Storage	Disk space for data, models, and logs	Minimum 2 GB free disk space
Software		
Frameworks	Python frameworks	Flask Web Framework
Libraries	Additional libraries	NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, Joblib / Pickle
Development Environment	IDE	Windows / Linux / macOS OS, Python 3.8+, Anaconda Navigator / Jupyter Notebook
Data		
Data	Source, size, format	Open-source Kaggle flood prediction dataset; tabular format featuring annual rainfall, cloud visibility, and seasonal features.